MATHEMATICS

University *of* NORTH ALABAMA

Fall 2025 Math Problem of the Week: Problem #8

Due noon Monday, October 20
Open to all UNA undergraduates

Directions: Deliver your written solution to me (Dr. Bryant) in CPTMTH 328 by noon Monday, October 20 or, if you cannot find me, you may leave it with Ms. Phifer in CPTMTH 377. All work must be your own. You are required to justify your response fully. If you utilize a well-known theorem, state the theorem's full name and proposition, showing that the conditions of the theorem are met. Good luck!

Problem #8: Black is to make 19 consecutive moves to arrive at such a position that white has checkmate in one move. No intermediate move can put white in check. How many distinct sequences of moves can black make to solve this puzzle?

